



Bivariate Data Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

- 1 The cost of making a hat is linearly related to the number of hats sold. To predict the costs, data was collected on the cost of making a hat for different levels of demands. The data is shown

below.

Number (hats)	10	15	20	25	30	35	40	45	50	55	60
Cost (dollars)	38	54	59	82	98	98	114	108	138	134	144

- (i) Find the equation of the least-squares line of best fit. Answer correct to one decimal place. 2

$A = 23.5$ and $B = 2.1$ (Statistics mode on calculator)

$y = mx + c = Bx + A$

place.

$\therefore$ Equation of the least-squares line of best fit is

$cost = 2.1 \times number + 23.5$

$cost = 2.1 \times number + 23.5$

$= 2.1 \times 36 + 23.5$

$= 99.1$

- (ii) What is the predicted cost of making 36 hats? 1

$\therefore$ Cost is \$99.10

- 2 A football coach is studying the relationship between the weight in kilograms of players and their vertical jump in metres. The coach believes that players who weigh less are able to jump higher. The data from ten players are shown in the table below.

weight w	84.5	83.3	58.7	74.1	70.7	72.3	75.8	71.8	63.2	65.9
vertical jump v	21.9	20.5	20.8	19.9	20.6	18.8	22.8	20.6	20.0	21.8

- (i) Find the equation of the least-squares regression line. Give each coefficient correct to two decimal places. 2

$$v \approx 0.03w + 18.75$$

- (ii) By calculating Pearson's correlation coefficient, correct to three decimal places, justify 2

whether the data confirms the association between weight and vertical jump.

$r \approx 0.199$ (3 decimal places)

This is a weak positive correlation. The data does not confirm that there is an association between bodyweight and vertical jump.

- 3 In a chess tournament, each player has a 'chess rating', which is a value from 1600 to 2000. Danny wants to know if there is a correlation between the number of years a player has spent playing chess and their chess rating. Danny randomly surveyed 10 players at the tournament. The results are shown in the

	Years spent playing chess (x)	Chess rating (y)
Player 1	1	1650
Player 2	2	1550
Player 3	2	1600
Player 4	3	1500
Player 5	4	1900
Player 6	4	1650
Player 7	6	1700
Player 8	8	1950
Player 9	10	1950
Player 10	15	2000

table. Beth, a competitor in the tournament, has spent 5 years playing chess. Use the equation of the least-squares regression line for this data set to determine Beth's chess rating. Give your answer correct to the nearest whole number.

$$y = 34.64x + 1554.49$$

$$\text{When } x = 5: y = 34.64 \times 5 + 1554.49 = 1727.69$$

Therefore, Beth's chess rating is predicted to be 1728.

- 4 A group of students were surveyed to find whether there was a correlation between a student's assessment marks out of 100 for Mathematics and Science. The results are in the table below.

Mathematics (x)	81	61	94	69	46	86
Science (y)	57	60	78	72	54	70

- (i) Find the correlation coefficient to 2 decimal places. $r \approx 0.72$ 1
- (ii) Find the equation of the least-squares regression line $y = mx + b$. You may find the values of m and b to 2 decimal places. $y = 0.39x + 37.06$ 1
- (iii) Using the equation, find the Science assessment mark for a student who scores 60 in Mathematics. $y = 0.39 \times 60 + 37.06 = 60.46$ 1
- (iv) A teacher extrapolates the data using the equation of the least-squares regression line to predict the assessment mark for Mathematics for a student who achieves 100 in Science. Justify using calculations why this is not possible. 1

$$100 = 0.39x + 37.06$$

$$x = 161.38... \approx 161.4$$

A student cannot obtain 161.4 as a mark as the maximum is 100.

- 5 Paolo and Gianluca are watching Italy in the UEFA Euro 2020 Championship Final. It is such an exciting match that Paolo is worried that Gianluca will faint with excitement. He takes his pulse at various intervals during the game. The information is recorded in the table and graphed on the scatterplot below.

t – time since the match started (mins)	2	5	8	16	20	30	42	48	53
b – heart rate (beats / min)	72	80	68	72	80	84	77	90	99

- (i) Find the value of r , the correlation coefficient, correct to 2 decimal places. $r = 0.80$ 1
- (ii) Find the equation of the least-squares regression line that can be used to predict b from t . Write the gradient and y -intercept correct to 2 decimal places, using the variables given in the table. 2
- (iii) What is Gianluca's predicted heart rate 46 minutes into the match? 1
- (iv) The match ended after 150 minutes. Paolo approximated Gianluca's heart rate to be 130 beats/min at the conclusion of the match. Is this a valid conclusion? Explain. 1

time (t) since the match started
only ranges from 2 mins to 53 mins.
150 mins is outside this range.
We should not extrapolate beyond
the given data set.
(It is not reasonable for his
heart rate to be that high
at 150 mins at the end of the match)

- 6 Ten students were ranked on their computer gaming ability on a new game. Each student also calculated the number of hours that they have played the game. The results are recorded in the table below.

Rank	1	2	3	4	5	6	7	8	9	10
Hours Played	198	143	88	102	82	94	54	36	20	12

- (i) Draw the scatterplot for the data in the table.



- (ii) Calculate, to 2 decimal places, Pearson's correlation coefficient, r , and describe the relationship between a player's rank and the number of hours that they have played.

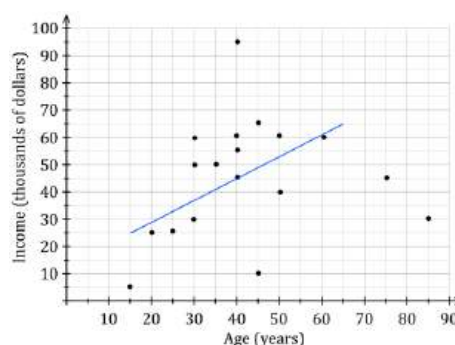
$r = -0.94$ There is a very strong negative relationship between rank and gaming hours. As time played increases, rank decreases.

- (iii) Find the equation of the least-squares regression line for the data given above.

Give your answer correct to 2 decimal places. $y = -17.74x + 180.47$

- 7 A sample of 18 randomly selected people were asked to state their Age (in years) and their Income (in \$1000).

The results are displayed in the scatterplot below with a line of best fit drawn.



- (i) Calculate the gradient of the line of best fit shown.

$$m = \frac{\text{Rise}}{\text{Run}} = \frac{40}{50} = 0.8$$

- (ii) What is the equation of the line of best fit shown?

$$y = mx + c$$

$$\text{Income} = 0.8 \times \text{Age} + c$$

To find c the line of best fit passes through (65, 65)

$$65 = 0.8 \times 65 + c \quad c = 13$$

$$\therefore \text{Income} = 0.8 \times \text{Age} + 13$$

- (iii) Use your equation to calculate the expected income of an 80 year old person.

$$\text{Income} = 0.8 \times \text{Age} + 13 = 0.8 \times 80 + 13 = 77$$

$\therefore$ Income of an 80 year old person is \$77 000.

- (iv) Comment on your result in part (iii).

People over the age of 65 often have a reduced income due to retirement. The prediction in part (c) is not accurate.

- (v) Describe the association between age and income.

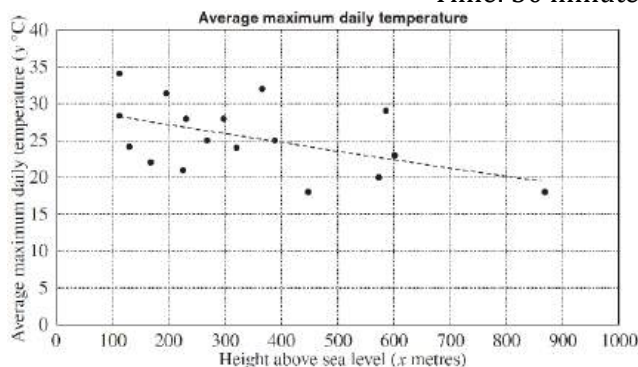
Association is linear, positive and weak.

Bivariate Data Topic Test 2 Solution

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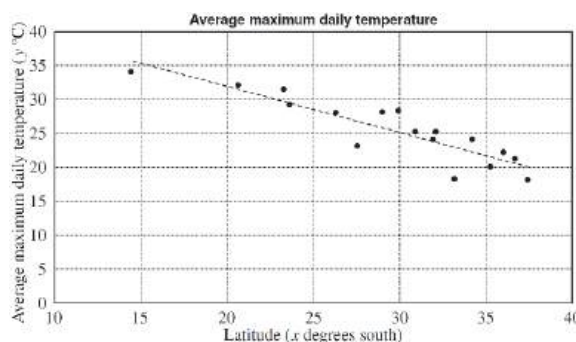
- 1 For a sample of 17 inland towns in Australia, the height above sea level, x (metres), and the average maximum daily temperature, y ($^{\circ}\text{C}$), were recorded. The graph shows the data as well as a regression line. The equation of the regression line is $y = 29.2 - 0.011x$. The correlation coefficient is $r = -0.494$.



- (a) (i) By using the equation of the regression line, predict the average maximum daily temperature, in degrees Celsius, for a town that is 540 m above sea level. Give your answer correct to one decimal place. $y = 29.2 - 0.011(540) \approx 23.3^{\circ}\text{C}$ (1 d.p.) 1
- (ii) The gradient of the regression line is -0.011 . Interpret the value of this gradient in the given context. 2

As the height of the town above sea level increases, the average maximum daily temperature decreases. Specifically, for every metre above sea level, there is approximately a 0.011°C decrease in the average maximum daily temperature. As the correlation coefficient is 0.494 , there is a moderate negative correlation of this data.

- (b) The graph below shows the relationship between the latitude, x (degrees south), and the average maximum daily temperature, y ($^{\circ}\text{C}$), for the same 17 towns, as well as a regression line.



The equation of the regression line is $y = 45.6 - 0.683x$. The correlation coefficient is $r = -0.897$. Another inland town in Australia is 540 m above sea level. Its latitude is 28 degrees south. Which measurement, height above sea level or latitude, would be better to use to predict this town's average maximum daily temperature? Give a reason for your answer.

Latitude would be a better predictor of the town's average daily maximum temperature as there is a stronger negative correlation for this data of $r = -0.897$ as opposed to only a moderate negative correlation coefficient of -0.494 given by the comparison of the height above sea level and average daily maximum temperature.

- 2 Luke suspects that the rate at which he spends cash is affected by the amount of cash he withdrew at his previous visit to an ATM. The table shows the amount of cash withdrawn, \$ x , from an ATM, and the time, y hours, until Luke's next withdrawal from an ATM.

Withdrawal	1	2	3	4	5	6	7	8	9	10
x	40	10	100	110	120	150	20	90	80	130
y	56	62	195	330	94	270	48	196	214	286

- (i) Find the equation of the least squares regression line for y in terms of x , for the withdrawals 1 to 10 and hence estimate how much cash (to the nearest \$10) Luke would need to withdraw from the ATM at his previous visit in order to not need to visit an ATM again for 120 hours. 2

from calculator: $y = A + Bx$
 $A = 30.26$ $B = 1.7$
 $y = 1.7x + 30.26$
 Find x at $y = 120$
 $120 = 1.7x + 30.26$
 $\therefore x = 52.79$
 $\therefore$ withdrawal = \$60

- (ii) Calculate the correlation coefficient between x and y for the withdrawals 1 to 10. 2

Describe the nature of the correlation. $r = 0.777$ {strong} positive correlation.

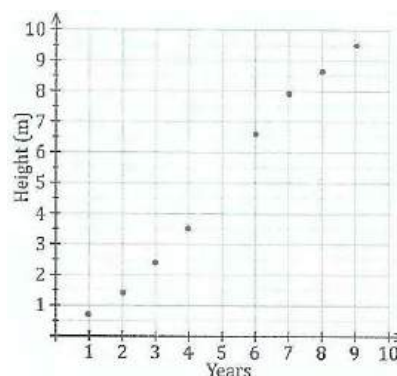
- 3 Tim is a scientist studying the growth of a particular tree over several years. The data he recorded is shown in the table below.

Years since planting, t	1	2	3	4	6	7	8	9
Height of tree, H metres	0.7	1.4	2.4	3.5	6.6	7.9	8.7	9.5

A scatterplot of the data is shown.

- (i) What is the Pearson's correlation coefficient? Answer correct to 4 decimal places. 1

- (ii) Find the equation of the least-squares line of best fit in terms of years (t) and height (h). Answer correct to 2 decimal places. 1



- (iii) Tim did not record the tree's height after five years. Predict the height after five years, correct to 1 decimal place. 1

- (iv) Estimate how many years it will take for the tree to reach a height of 24 metres. Answer correct to 1 decimal place. 1

$\therefore h = 1.19t - 0.85$
 when $t = 5$, $h = 1.19(5) - 0.85 = 5.1m$

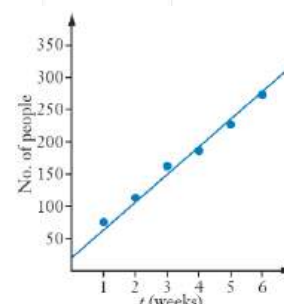
- 4 This table shows the results of a survey into the number of people who attend a new restaurant over several weeks.

t (weeks)	1	2	3	4	5	6
No. of people	76	114	163	187	228	274

- (i) Draw a scatterplot and sketch a line of best fit. 2

- (ii) Find the equation of this line. 2

$$N = 37.5t + 42.5$$

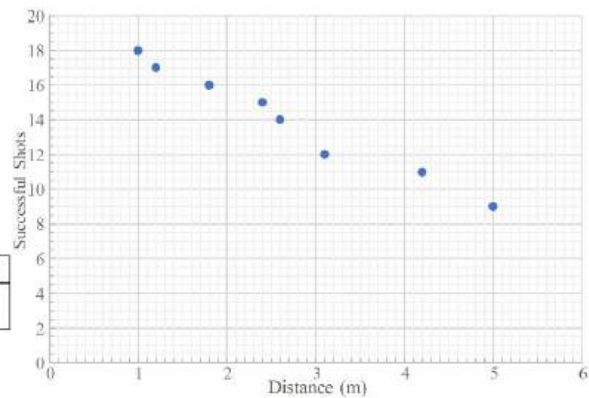


(iii) Use the equation to find the number of people you would expect to attend the restaurant after 10 weeks. **418** 1

(iv) Is this equation a good model for the number of people attending the restaurant? 1
No, because the number of people cannot keep increasing forever.

5 A netball player took 20 shots from each of 8 different positions that were different distances from the goal. The results are shown in the table below.

Distance (m)	1.2	2.4	5.0	4.2	1.0	1.8	3.1	2.6
Successful Shots	17	15	9	11	18	16	12	14



(i) Using the scatterplot, describe the correlation between the two variables. Very strong negative linear correlation 1

(ii) Calculate Pearson's correlation coefficient for this data, correct to 3 decimal places. 1
 $r = -0.988$

(iii) Determine the equation of the least-squares regression line for this data, rounding any values to 2 decimal places. **$y = 19.85 - 2.198x$**
Successful shots = $19.85 - 2.198 \times \text{Distance}$ 1

(iv) Using your equation from part (c), predict the distance, to the nearest cm, at which the netball player will make 10 successful shots. 2
 $10 = 19.85 - 2.198 \times \text{Distance}$
 $\text{Distance} = \frac{10 - 19.85}{-2.198} = \boxed{4.48 \text{ m}}$

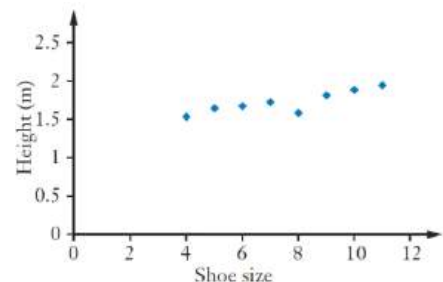
6 The table shows heights and shoe sizes of several males.

Shoe size	4	5	6	7	8	9	10	11
Height (m)	1.54	1.65	1.68	1.73	1.59	1.82	1.89	1.95

(i) Draw a scatterplot to show this data. 1

(ii) Find the correlation coefficient. **0.876** 1

(iii) Find the equation of the least-squares regression line. **$y = 0.052x + 1.34$** 1



(iv) The male in the sample with shoe size 8 is found to be an outlier for this data. For the sample without this outlier, find:

(a) the correlation coefficient **0.992** 1

(b) the equation of the least-squares regression line **$y = 0.054x + 1.35$** 1

(v) Is this equation a good model for shoe sizes and height? 1
No. Shoe sizes cannot keep increasing